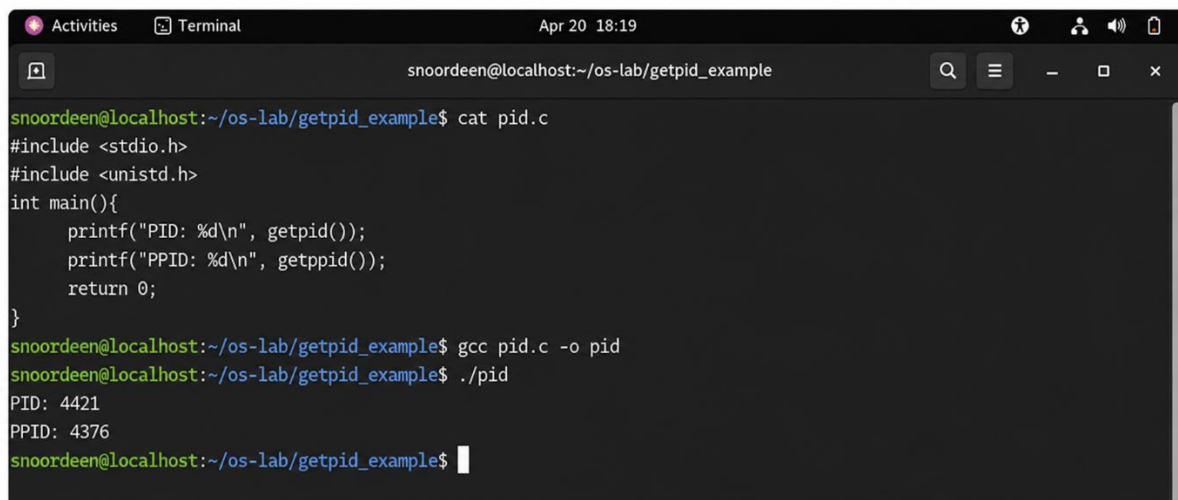


OPERATING SYSTEM CONCEPT – LAB

EXP -04 SYSTEM CALLS

getpid()



```
Activities Terminal Apr 20 18:19
snoordeen@localhost:~/os-lab/getpid_example

snoordeen@localhost:~/os-lab/getpid_example$ cat pid.c
#include <stdio.h>
#include <unistd.h>
int main(){
    printf("PID: %d\n", getpid());
    printf("PPID: %d\n", getppid());
    return 0;
}

snoordeen@localhost:~/os-lab/getpid_example$ gcc pid.c -o pid
snoordeen@localhost:~/os-lab/getpid_example$ ./pid
PID: 4421
PPID: 4376
snoordeen@localhost:~/os-lab/getpid_example$
```

readdir()

```
Activities Terminal Apr 20 19:28
snoordeen@localhost:~/os-lab/readdir_example$ cat readdir.c
#include <stdio.h>
#include <dirent.h>
int main(){
    DIR *dir;
    struct dirent *entry;
    dir=opendir(".");
    if(dir==NULL){
        printf("Cannot open directory!\n");
        return 1;
    }
    printf("Files in directory:\n");
    while((entry=readdir(dir))!=NULL){
        printf("%s\n", entry->d_name);
    }
    closedir(dir);
    return 0;
}
snoordeen@localhost:~/os-lab/readdir_example$ gcc readdir.c -o readdir
snoordeen@localhost:~/os-lab/readdir_example$ ./readdir
Files in directory:
.
..
pid.c
opendir.c
readdir.c
fork.c
fork
exec.c
exec
exec
pid
opendir
readdir
snoordeen@localhost:~/os-lab/readdir_example$
```

fork()

```
Activities Terminal Apr 20 19:31
snoordeen@localhost:~/os-lab/fork_example$ cat fork.c
#include <stdio.h>
#include <unistd.h>
int main(){
    pid_t pid=fork();
    if(pid==0){
        printf("This is CHILD process\n");
    } else {
        printf("This is PARENT process\n");
    }
    printf("PID: %d\n", getpid());
    return 0;
}
snoordeen@localhost:~/os-lab/fork_example$ gcc fork.c -o fork
snoordeen@localhost:~/os-lab/fork_example$ ./fork
This is PARENT process
PID: 4421
This is CHILD process
PID: 4422
snoordeen@localhost:~/os-lab/fork_example$
```

exec()

```
Activities Terminal Apr 20 19:32
snoordeen@localhost:~/os-lab/evex_example$ cat exec.c
#include <stdio.h>
#include <unistd.h>
int main(){
    printf("Before exec()\n");
    execl("/bin/ls", "ls", NULL);
    printf("After exec()\n");
    return 0;
}
snoordeen@localhost:~/os-lab/evex_example$ gcc exec.c -o exec
snoordeen@localhost:~/os-lab/evex_example$ ./exec
Before exec()
exec  exec.c  fork  fork.c  open_dir  open_dir.c  pid  pid.c  readdir  readdir.c
snoordeen@localhost:~/os-lab/evex_example$
```